

Right-angled triangles



1

a AB

b BC

2

a AC

b BC

c BC

d AC

e $\angle ACB$

3

a

i

$$\frac{4}{3}$$

ii

$$\frac{4}{5}$$

iii

$$\frac{3}{5}$$

b

i

$$\frac{15}{8}$$

ii

$$\frac{15}{17}$$

iii

$$\frac{8}{17}$$

c

i

$$\frac{21}{20}$$

ii

$$\frac{21}{29}$$

iii

$$\frac{20}{29}$$

d

i

$$\frac{12}{5}$$

ii

$$\frac{12}{13}$$

iii

$$\frac{5}{13}$$

e

i

$$\frac{24}{7}$$

ii

$$\frac{24}{25}$$

iii

$$\frac{7}{25}$$

f

i

$$\frac{28}{45}$$

ii

$$\frac{28}{53}$$

iii

$$\frac{45}{53}$$

Trigonometric ratios

4

a

i

$$\tan \theta = \frac{AC}{BC}$$

ii

$$\sin \alpha = \frac{BC}{AB}$$

b

i

$$\sin \theta = \frac{AC}{AB}$$

ii

$$\tan \alpha = \frac{BC}{AC}$$

c

i

$$\cos \theta = \frac{BC}{AB}$$

ii

$$\cos \alpha = \frac{AC}{AB}$$

d

i

$$\tan \theta = \frac{8}{15}$$

ii

$$\sin \alpha = \frac{15}{17}$$

e

i

$$\cos \theta = \frac{15}{17}$$

ii

$$\tan \alpha = \frac{15}{8}$$

f

i

$$\sin \theta = \frac{24}{25}$$

ii

$$\cos \alpha = \frac{24}{25}$$

5

a $x = 10$

b

$$\frac{4}{5}$$

c

$$\frac{3}{5}$$

6

a $x = 4$

b

$$\frac{4}{3}$$



Unknown sides

7 $b = 8$

8 $d = 3084$

9 $b = 19.8$

10 a $f = 4.08$

b $f = 6.12$

c $g = 9.46$

d $a = 11.59$

Unknown angles

11 $\frac{\sqrt{15}}{7}$

12 a $x = 67^\circ$

b $x = 74^\circ$

13 $A = 66.4^\circ$

14 $\tan \theta = \frac{24}{7}$

Applications

15 a $h = 21$ b $f = 22$ c $h = 98, x = 28$ d $x = 34$
e $x = 12$

16 $d = 1.48$

17 1.96 m

18 $y = 58.02$

19



20 $\theta = 9.88^\circ$

21 $y = 54.90^\circ$

22 $\tan \theta = \frac{4}{3}$

23 $h = 4.4 \text{ m}$